

Cluster-Wide Task Slowdown Detection in Cloud System

A DSA Research Presentation

the health of cloud systems directly controls user satisfaction and prevents massive financial penalties due to service contract violations (SLAs). We need an improved method to spot systemic failures.

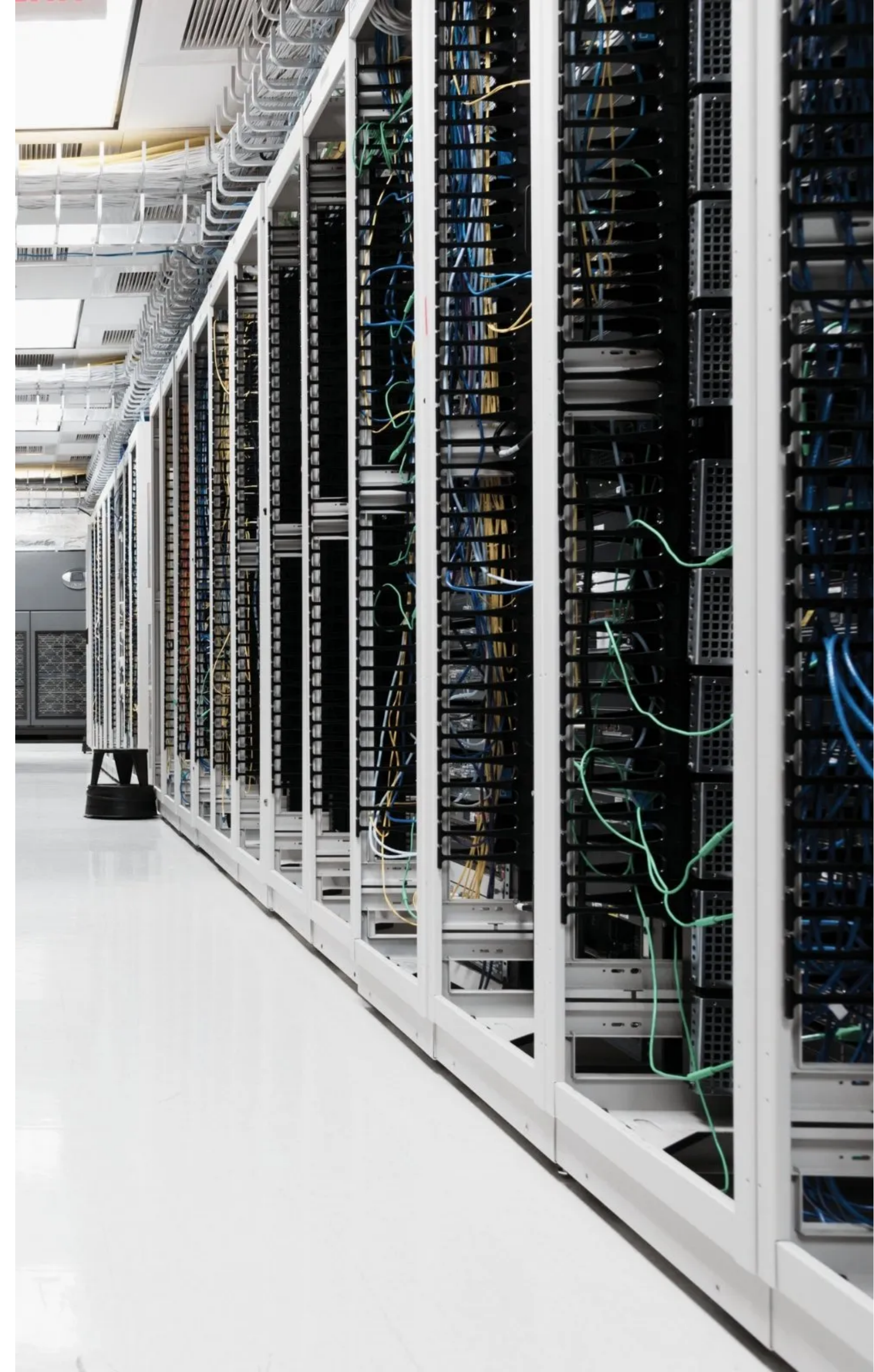
SLA violations are failures by a service provider to meet the terms of a Service Level Agreement (SLA), which can result in penalties.

The Core Problem

Inefficiency and Noise

- **The Cost of Individual Monitoring:**

Cloud companies run **millions of tasks** simultaneously. It is computationally **impossible and inefficient to track** every single one individually.

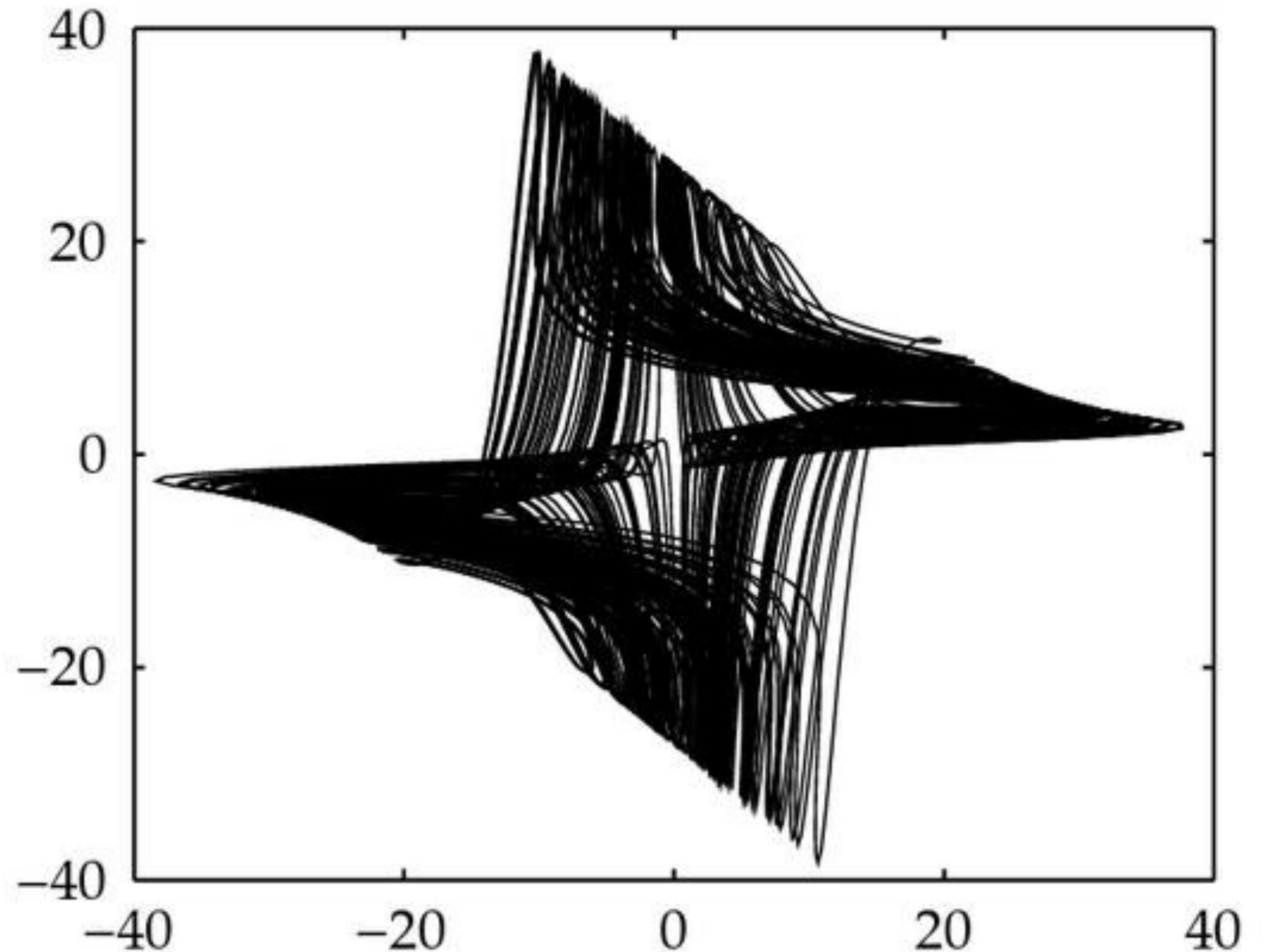


The Core Problem

Inefficiency and Noise

- **The Flaw of False Alarms (Noise):**

Most slowdowns are just noise. Tasks run in volatile, virtual environments, causing violent fluctuations in speed. A single task slowing down doesn't mean the whole cluster is failing. If we flag every one, we get overwhelming **false alarms** (false positives).

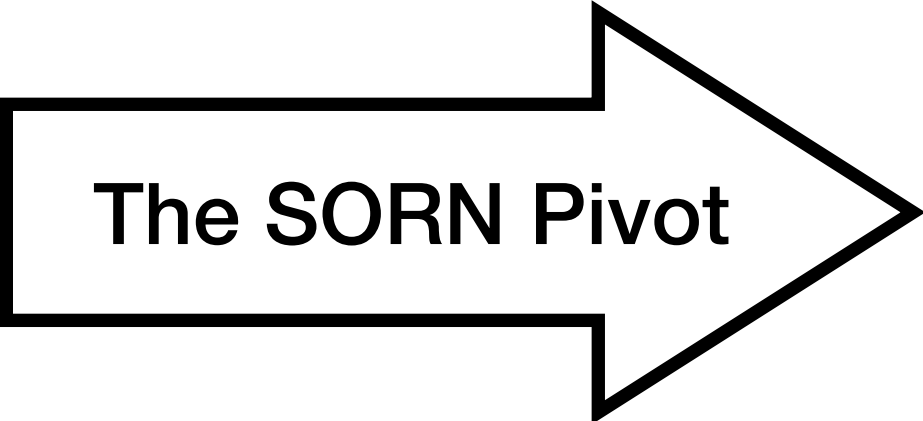


Violent Fluctuation Noise | False Positives

The Core Problem

The Goal

We need a new method that detects cluster-wide malfunctions by looking at patterns, not individual noise



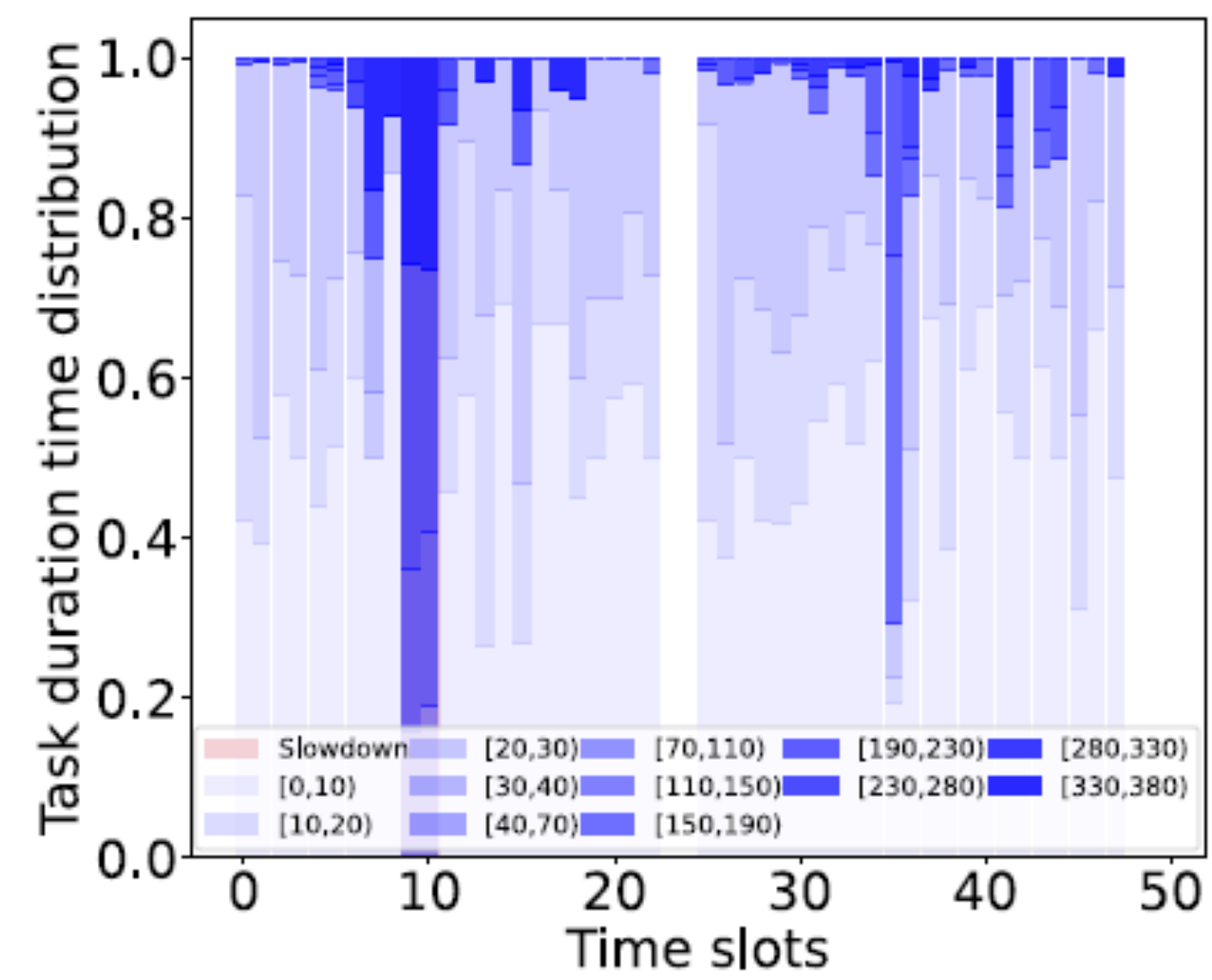
The SORN Pivot

The New Approach: Cluster-Wide Distribution

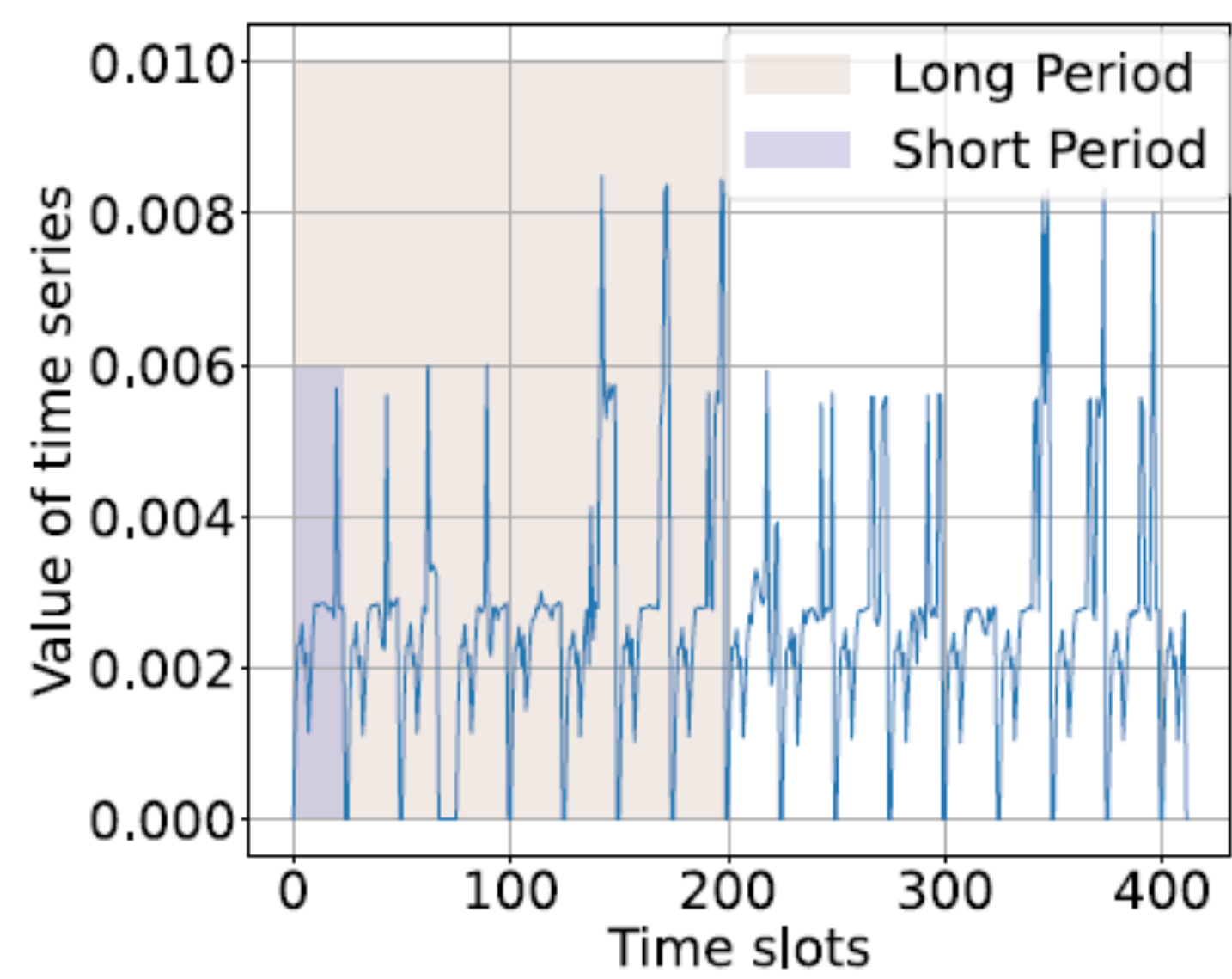
SORN

To address the chaos and massive scale of modern cloud systems, the researchers developed SORN.

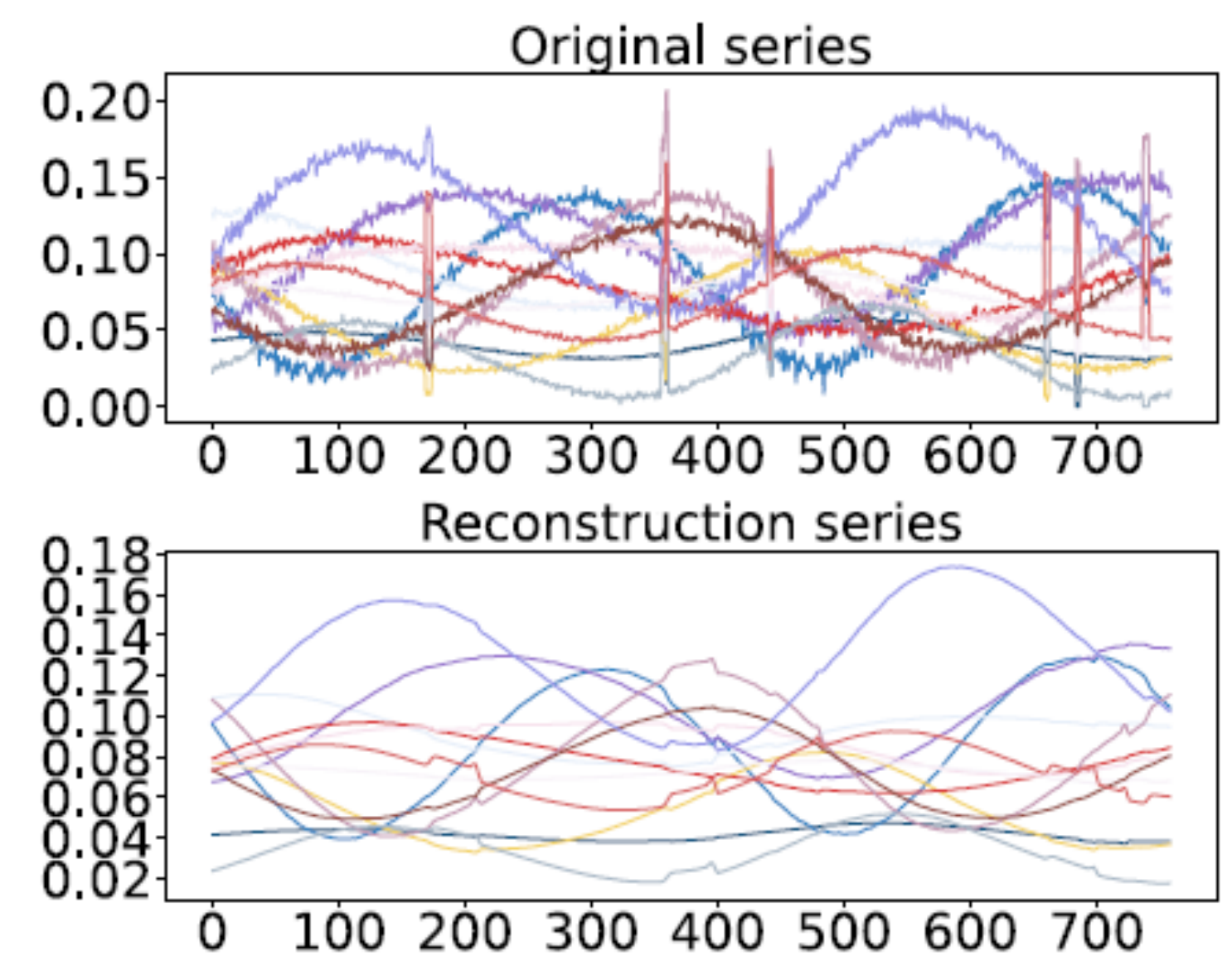
The name defines its purpose: *Skimming Off subperiods... and Reconstructing Non-slowing fluctuation*. It starts with a simple but powerful data structure transformation.



(a) The task duration time distribution



(b) The compound periodicity



(c) The reconstruction series of attention

The New Approach: SORN for Cluster-Wide Distribution

The New Approach: SORN for Cluster-Wide Distribution

SORN Defined

SORN is a deep learning framework. The name is an acronym defining its core function: **Skimming Off subperiods in descending amplitude order and Reconstructing Non-slowing fluctuation.**

The New Approach: SORN for Cluster-Wide Distribution

The Input Change

We stop using chaotic, individual task data. We start using the **Task Duration Time Distribution**, it's a summary of how fast all tasks are running at a given moment in time.

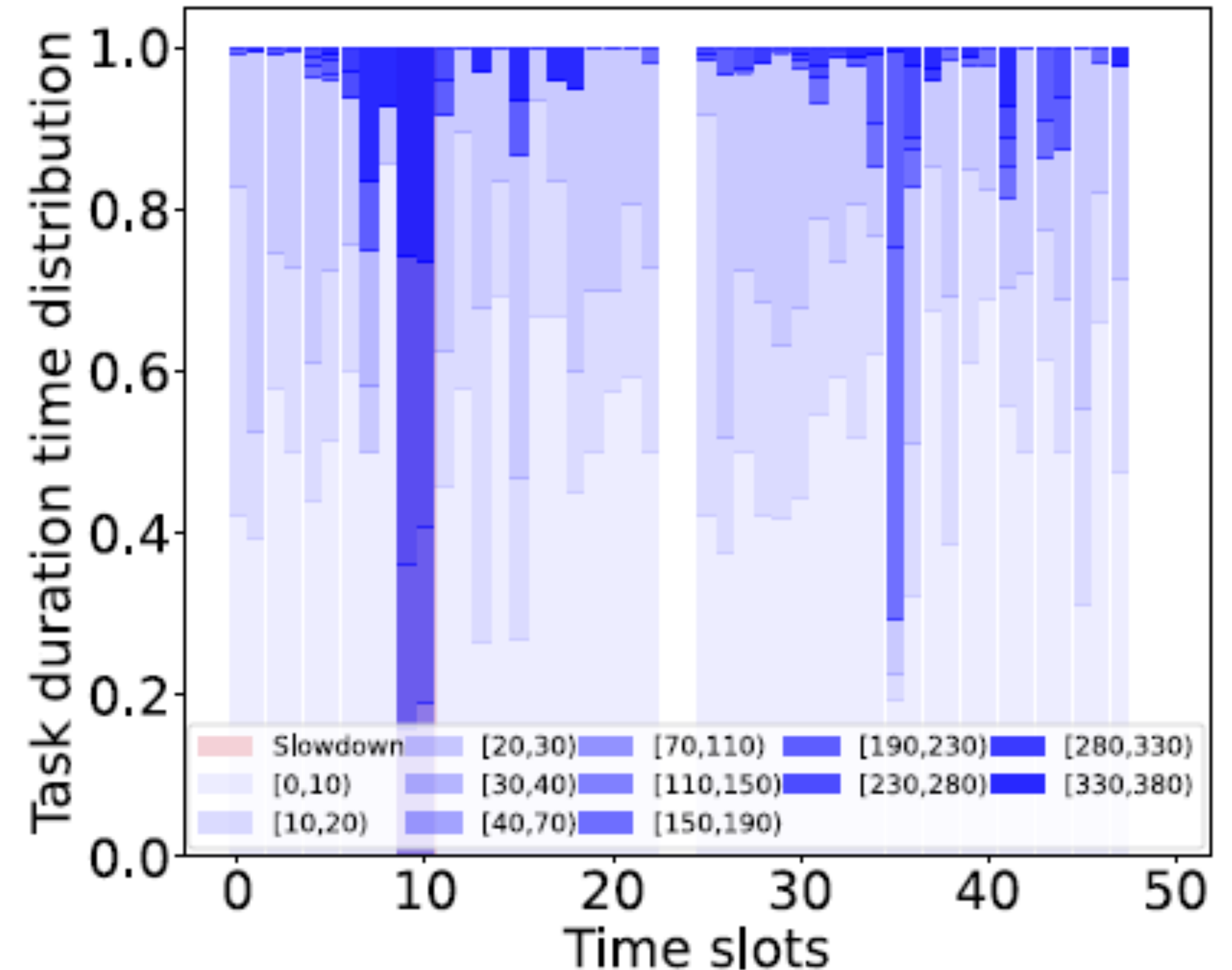
The New Approach: SORN for Cluster-Wide Distribution

Data Step 1: Partitioning

We divide the entire range of possible task times (e.g., 0 minutes up to 6 hours) into fixed speed intervals, such as those shown on the y-axis of the graph. (These intervals are our dimensions D).

Note: Data structure we created is a multivariate time series $x \in \mathbb{R}^{T \times D}$, where D is the number of speed intervals.

$$x \in \mathbb{R}^{T \times D}$$



(a) The task duration time distribution

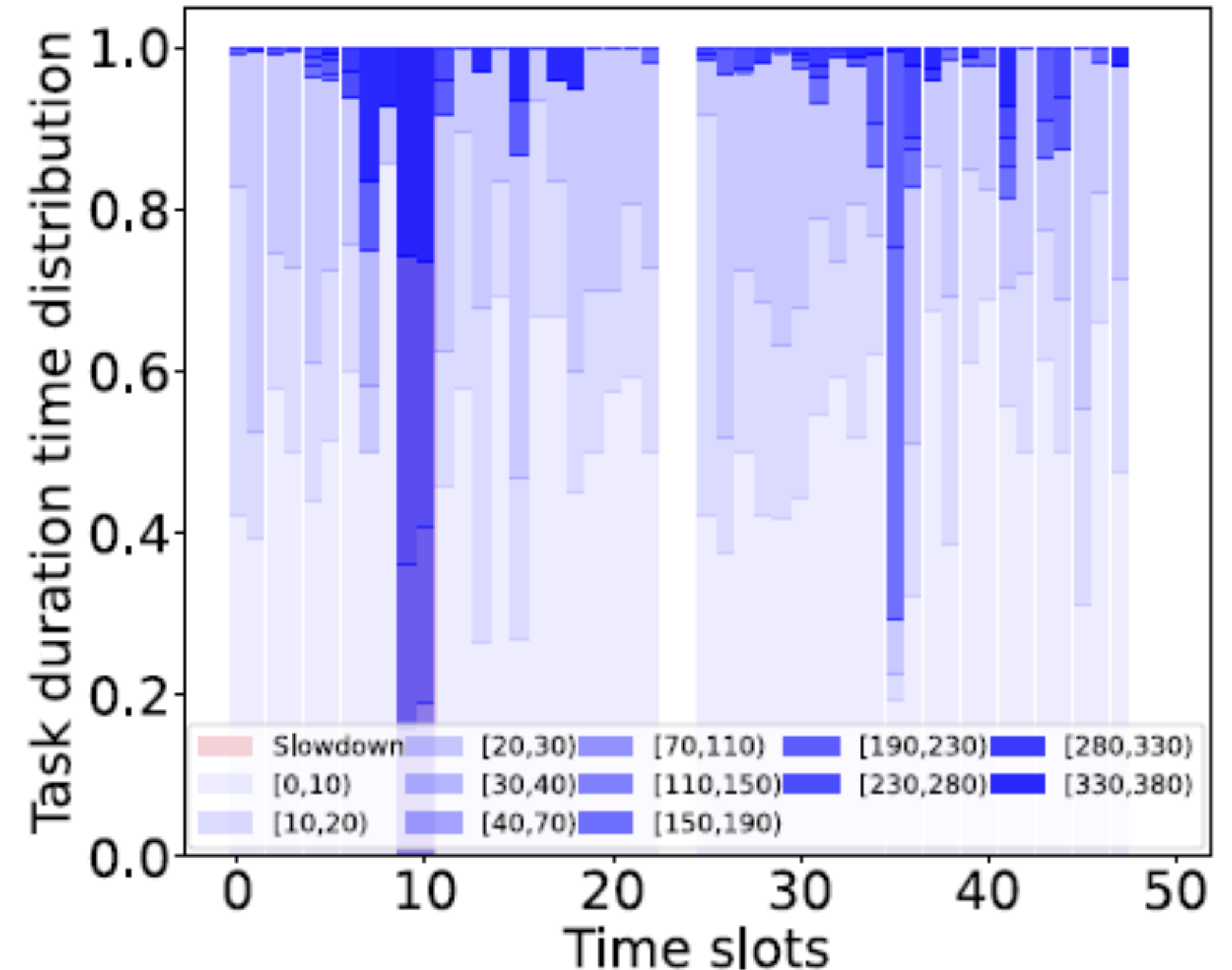
The New Approach: SORN for Cluster-Wide Distribution

Data Step 2: Counting

At every time slot t (x-axis), we simply **count the percentage** of tasks that fall into each speed interval (the colored blocks).

Note: Data structure we created is a multivariate time series $x \in \mathbb{R}^{T \times D}$, where D is the number of speed intervals.

$$x \in \mathbb{R}^{T \times D}$$



(a) The task duration time distribution

The New Approach: SORN for Cluster-Wide Distribution

The Result

This provides us with a **structured time series**. We gain two advantages:

- **Efficiency:** The algorithm scales, independent of task count.
- **Clarity:** The noise cancels out, revealing regular patterns.

The Three Major Technical Hurdles SORN Must Overcome

Now that we have clean, aggregate data, we face three sophisticated challenges unique to analyzing complex system patterns. SORN provides three dedicated, innovative modules to solve each problem.

The Three Major Technical Hurdles SORN Must Overcome

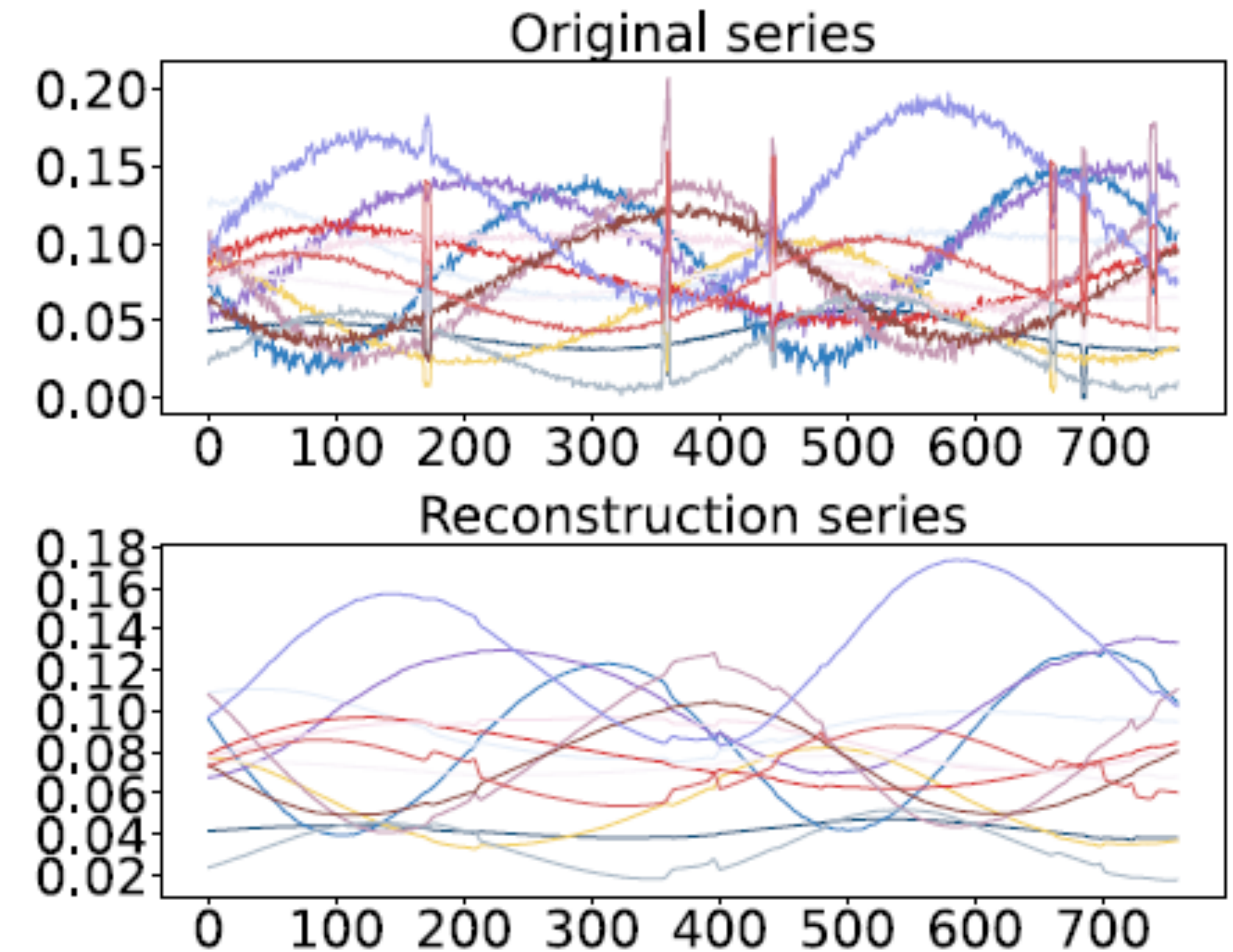
1. Compound Periodicity

Problem:

The cluster has many schedules (daily, weekly) running at once. Standard AI only captures the **loudest cycle**, ignoring the quieter, necessary patterns (low amplitude).

SORN's Solution:

Skimming Attention (SA): Peels off the cycles one by one to ensure all patterns are modeled.



(c) The reconstruction series of attention

The Three Major Technical Hurdles SORN Must Overcome

2. Non-Slowing Fluctuations

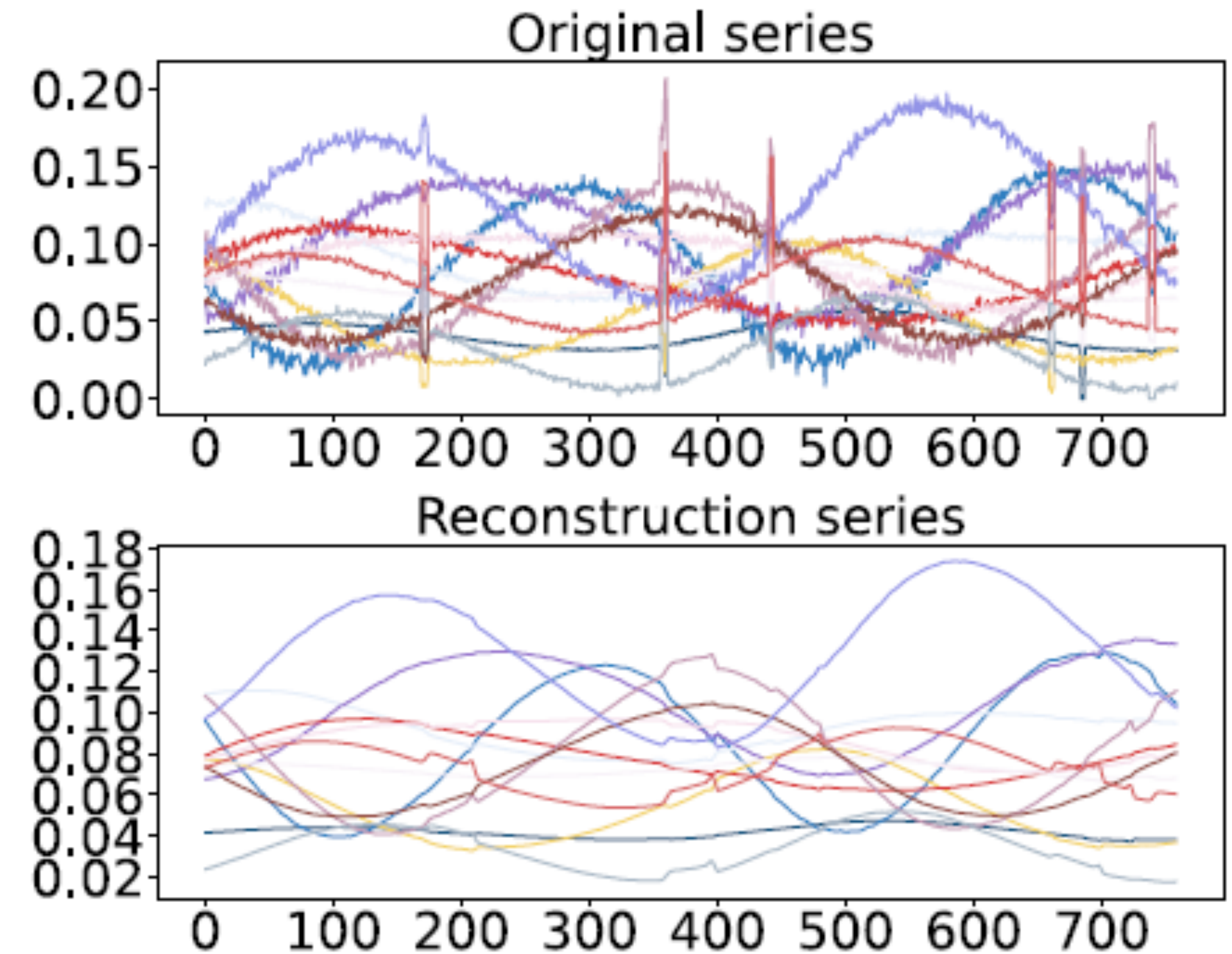
Problem:

We only want to detect **slowdowns (BAD)**. The system also has beneficial task **speedups (GOOD)**. The model must learn to ignore the speedups to avoid high false alarms.

SORN's Solution:

Neural Optimal Transport (OT):

Selectively filters out only the speedups, leaving only true slowdowns to trigger the alarm.



(c) The reconstruction series of attention

The Three Major Technical Hurdles SORN Must Overcome

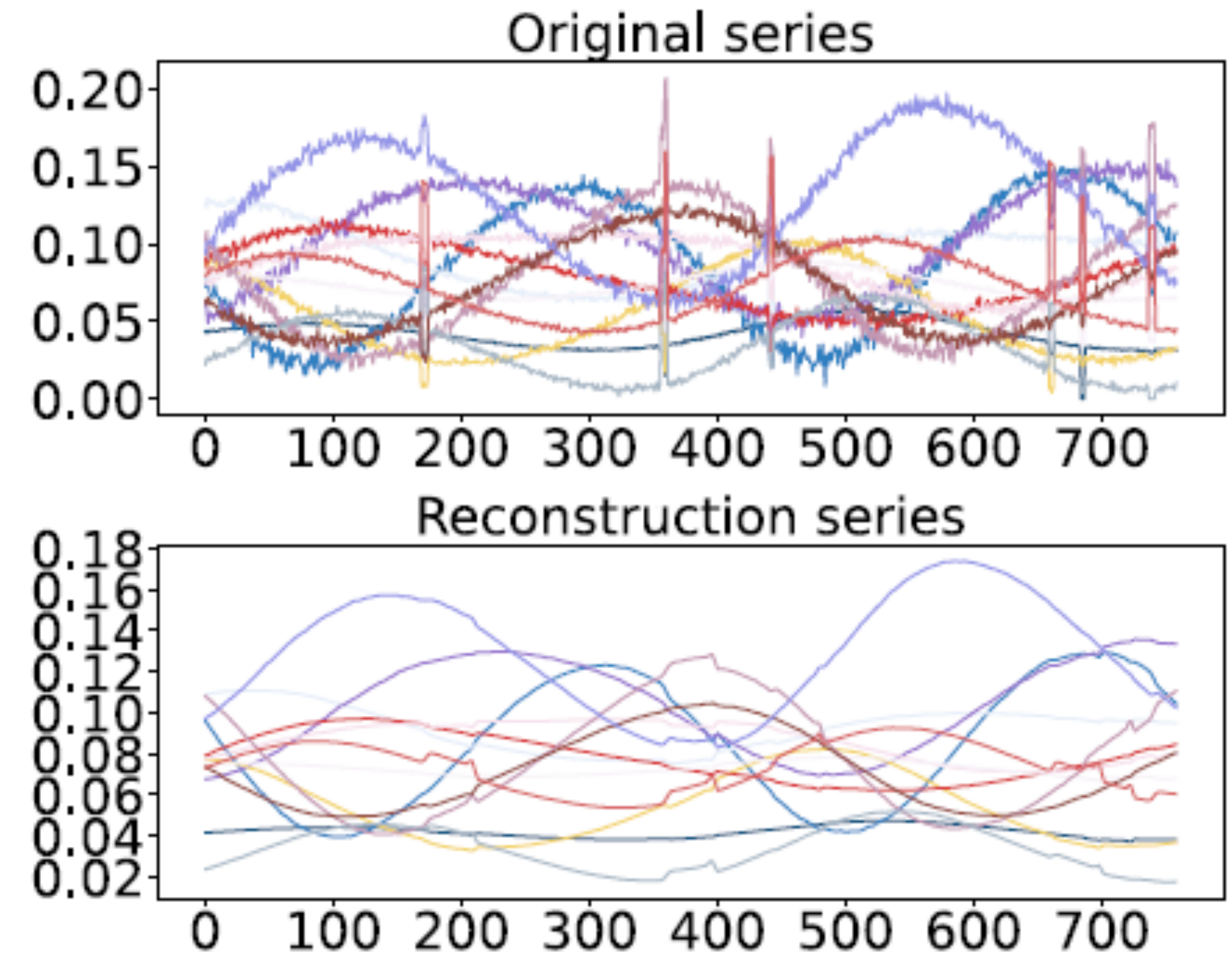
3. Anomalies in Training Data

Problem:

The model is trained on historical data that is often **contaminated** with real, unlabeled anomalies. If the model learns these mistakes, it fails to detect them later.

SORN's Solution:

Picky Loss Function: Gives higher importance (weight) to reliable, predictable data points during training.



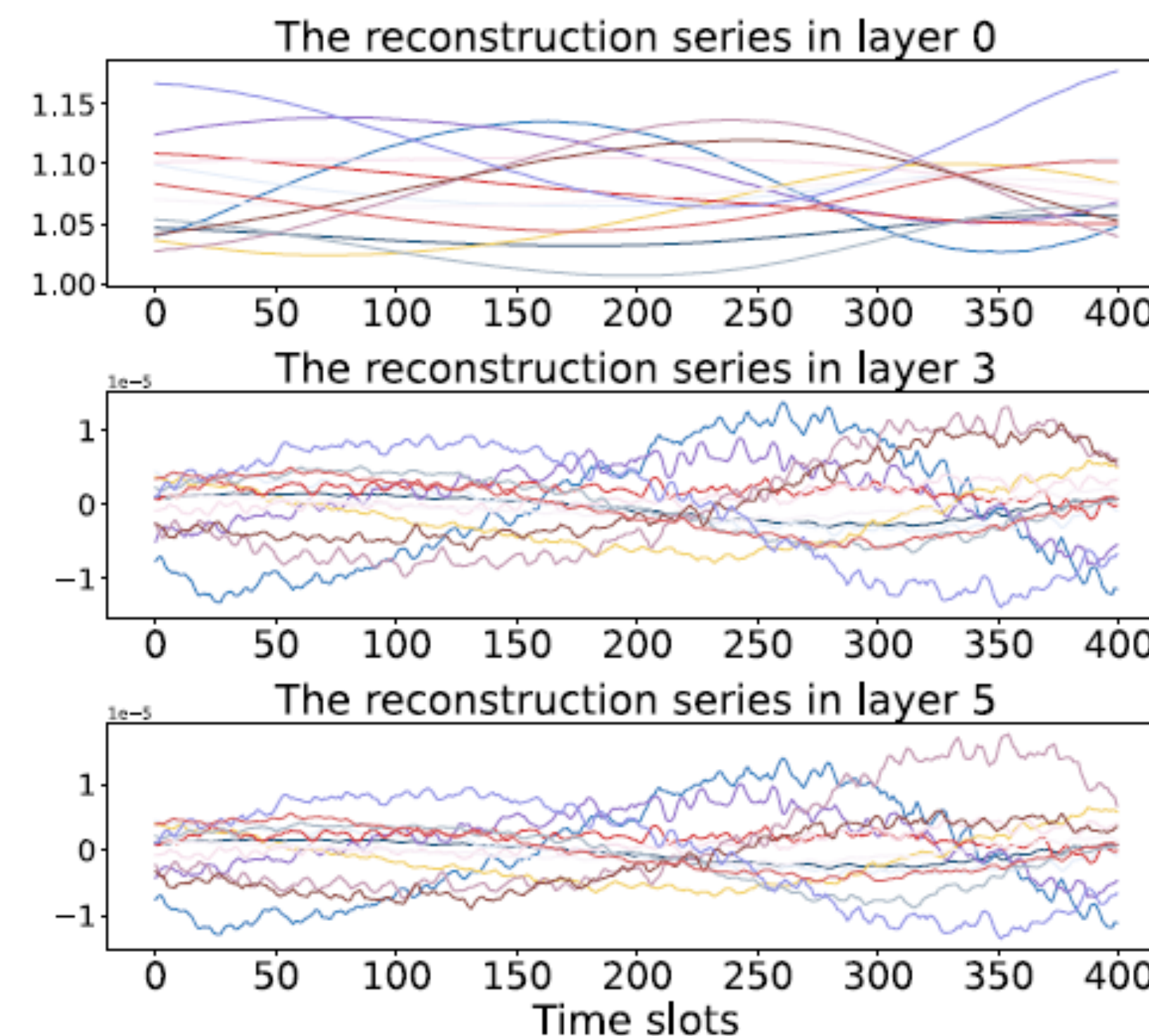
(c) The reconstruction series of attention

SORN's Solution 1: Skimming Attention (SA)

Goal: Master Compound Periodicity (Fixing the high-amplitude bias)

- **The Problem:**

Attention mechanisms pay more attention to subperiods with higher amplitudes (loud signals), which prevents accurate reconstruction of subtle, low-amplitude patterns.



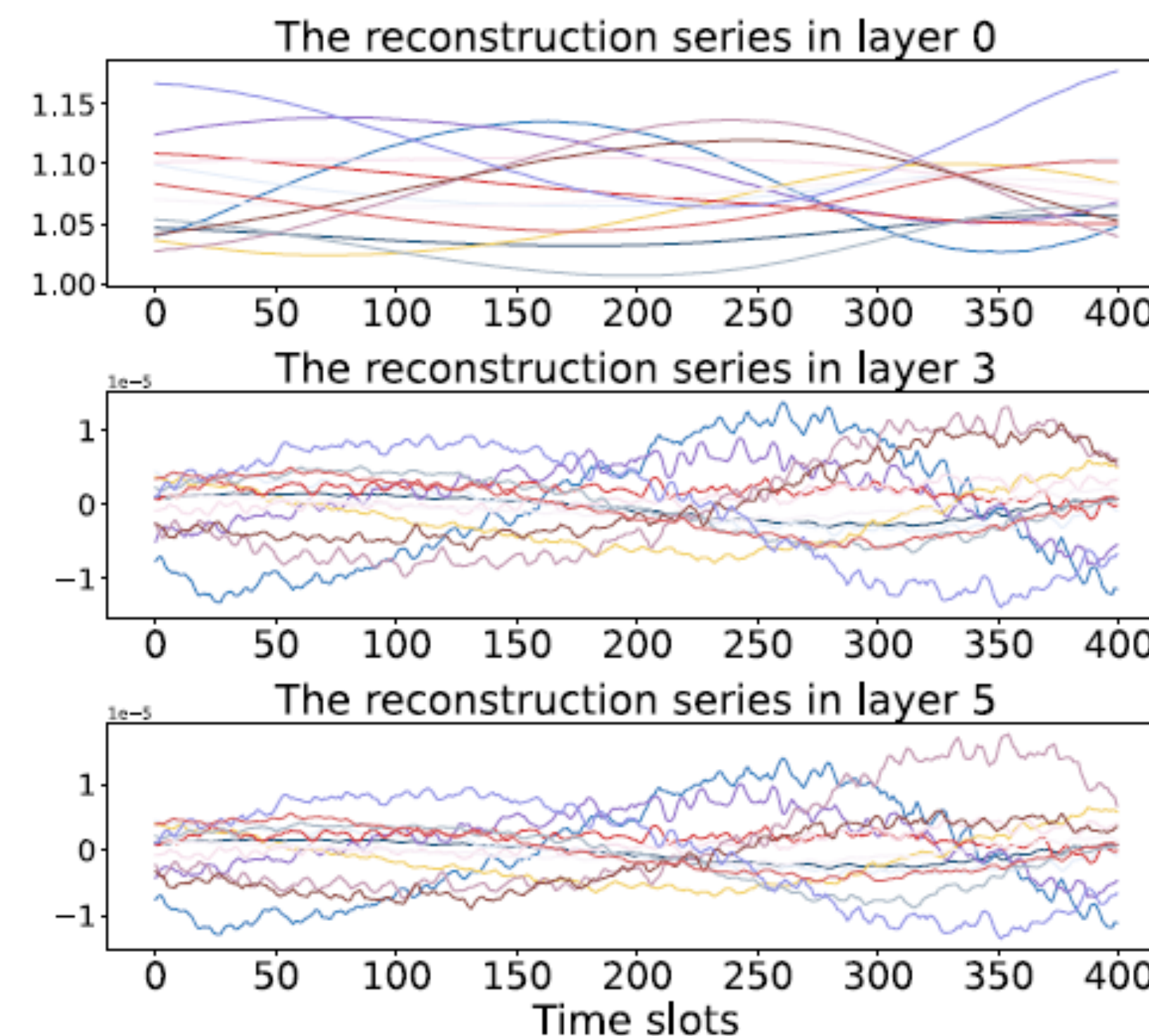
(c) The series from different Skimming Attention layers

SORN's Solution 1: Skimming Attention (SA)

Goal: Master Compound Periodicity (Fixing the high-amplitude bias)

- **The Solution:**

SORN uses a multi-layer **Skimming Attention (SA)** process that iteratively reconstructs and separates periodic components in descending amplitude order.



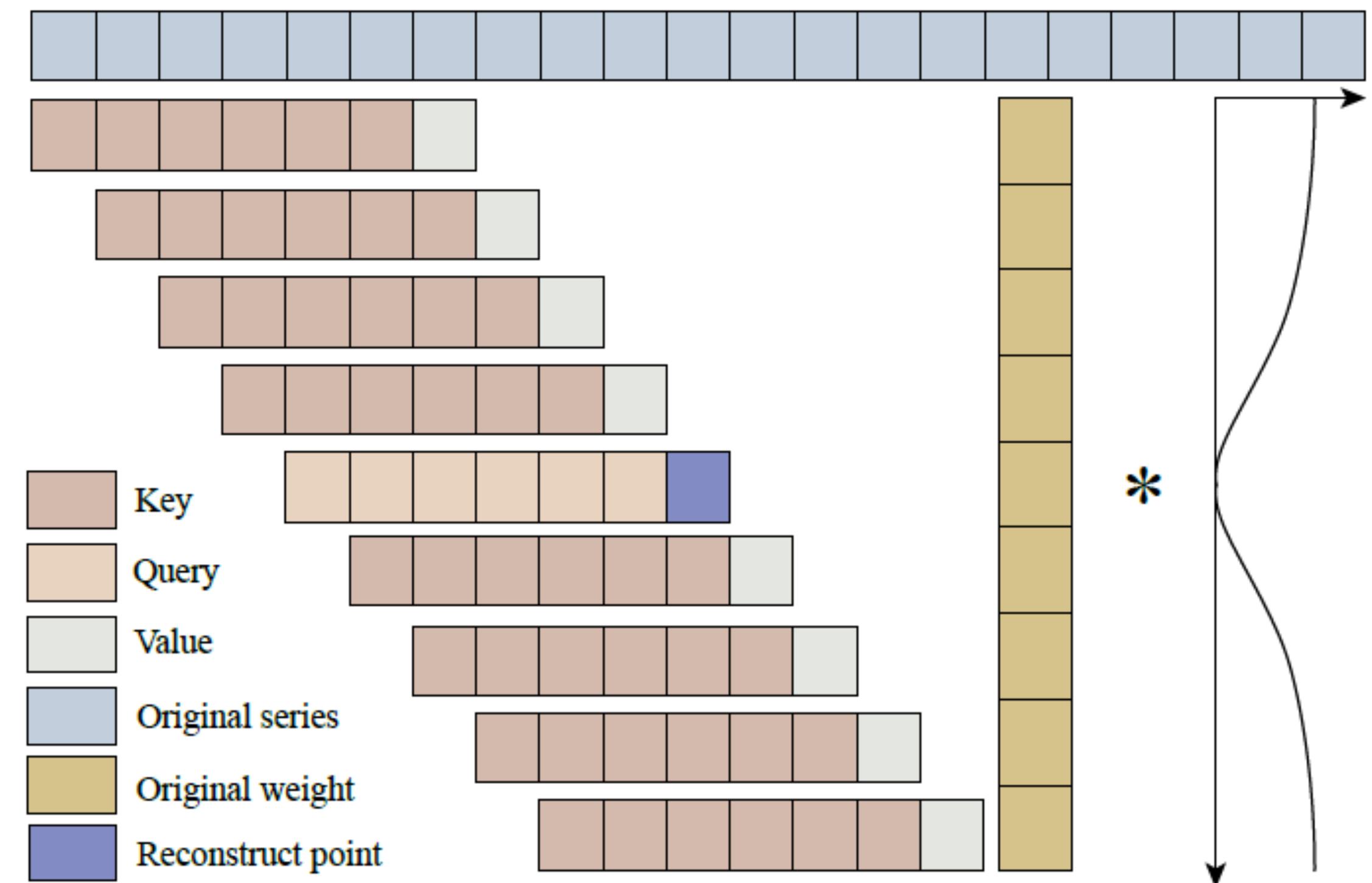
(c) The series from different Skimming Attention layers

SORN's Solution 1: Skimming Attention (SA)

Goal: Master Compound Periodicity (Fixing the high-amplitude bias)

- **How it Works:**

Each layer reconstructs and removes the current highest amplitude signal. The remaining residual signal then has a new highest-amplitude signal (the next quietest one), allowing the subsequent layer to focus on and successfully reconstruct it.



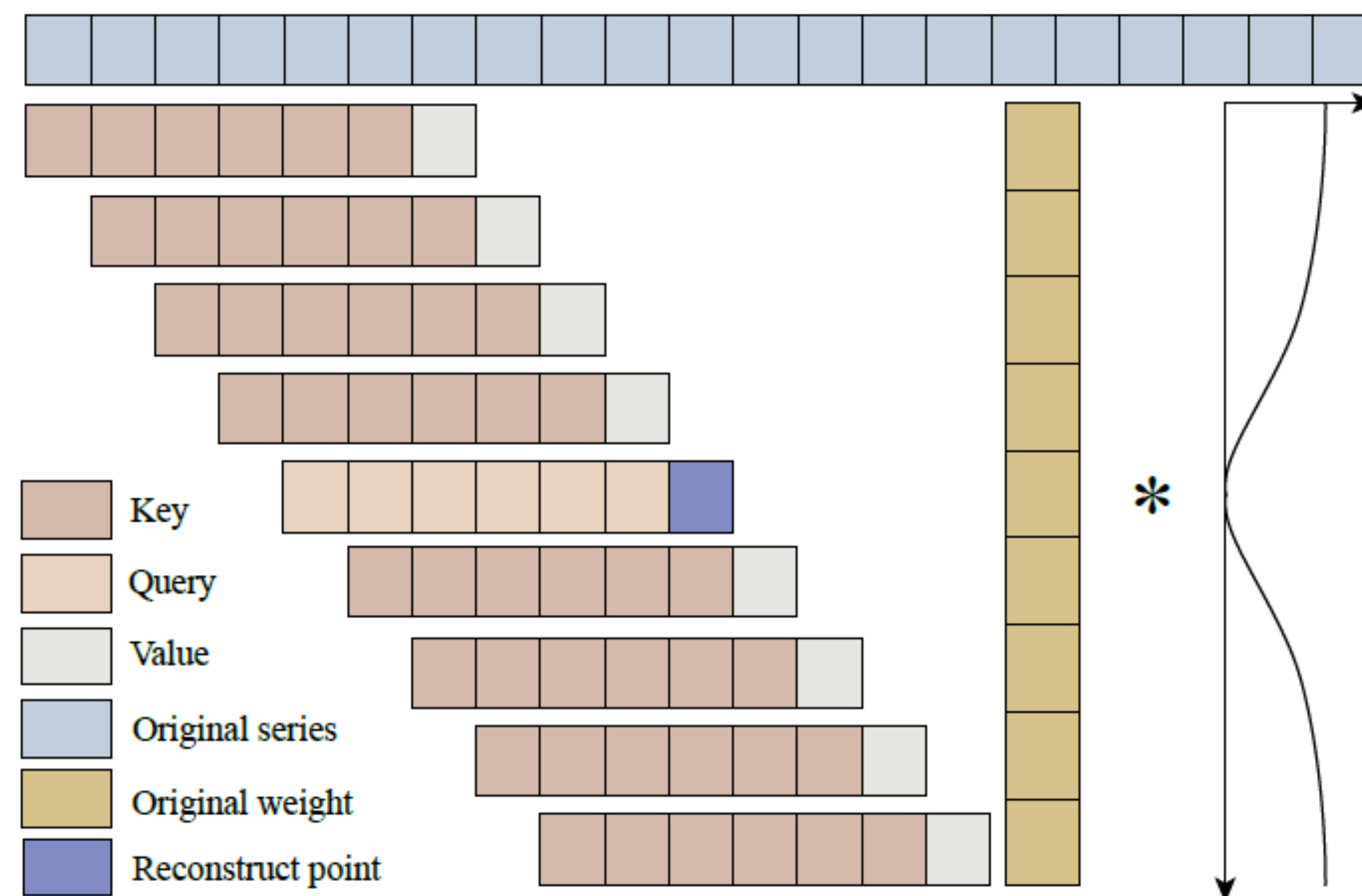
(b) A layer of Skimming Attention

SORN's Solution 1: Skimming Attention (SA)

Goal: Master Compound Periodicity (Fixing the high-amplitude bias)

- **Key Function:**

A special **Gate Curve** is used to ensure the attention mechanism prioritizes long-range, periodic correlations (e.g., *correlations between two Sundays*) over just local similarity.



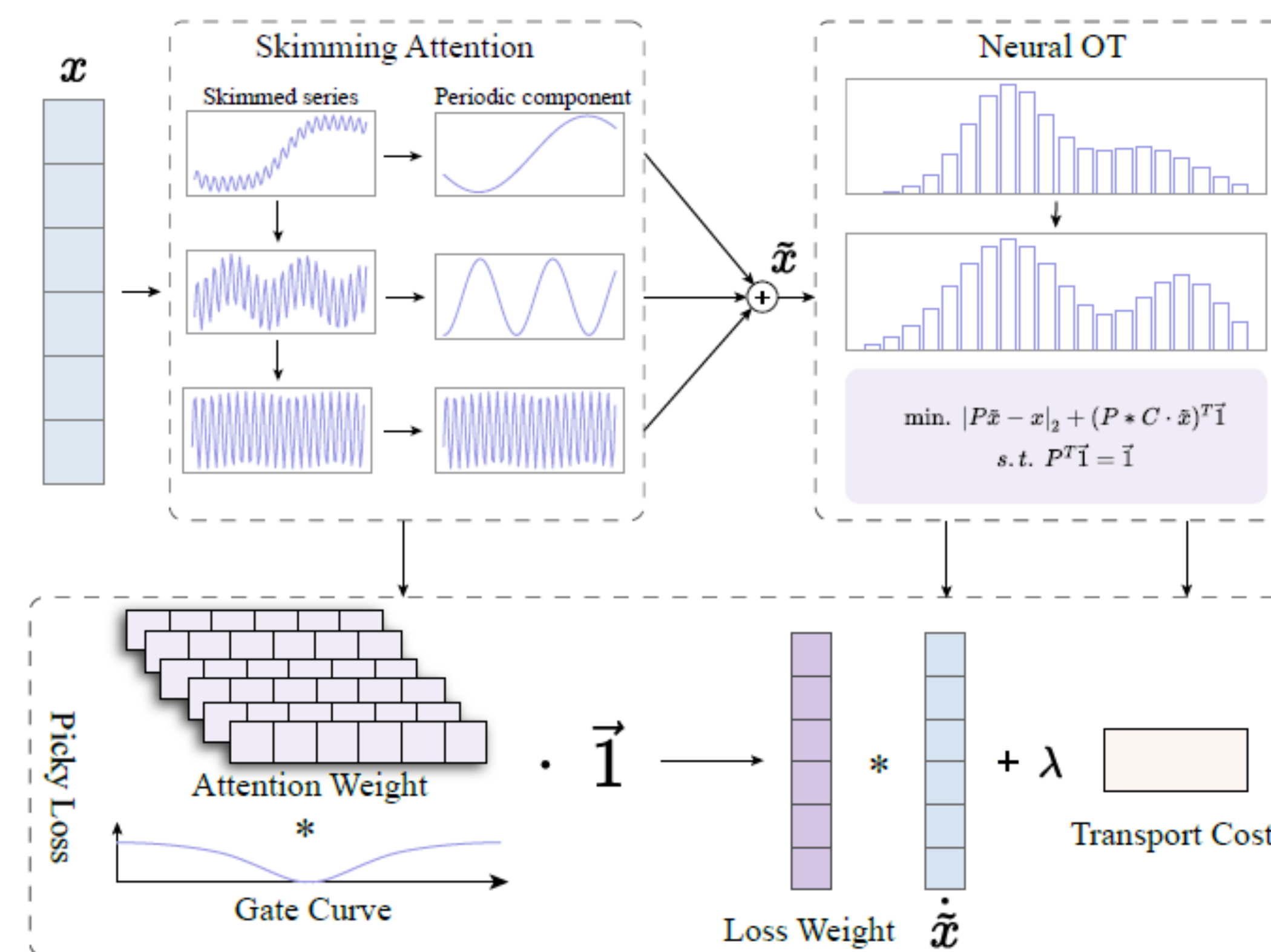
(b) A layer of Skimming Attention

SORN's Solution 2: Neural Optimal Transport (OT)

Goal: Filter Non-Slowing Fluctuations (Distinguishing slowdowns from speedups)

- The Problem:**

The model must successfully reconstruct all non-slowdown events, such as task speedups, so they are not mistaken for anomalies.



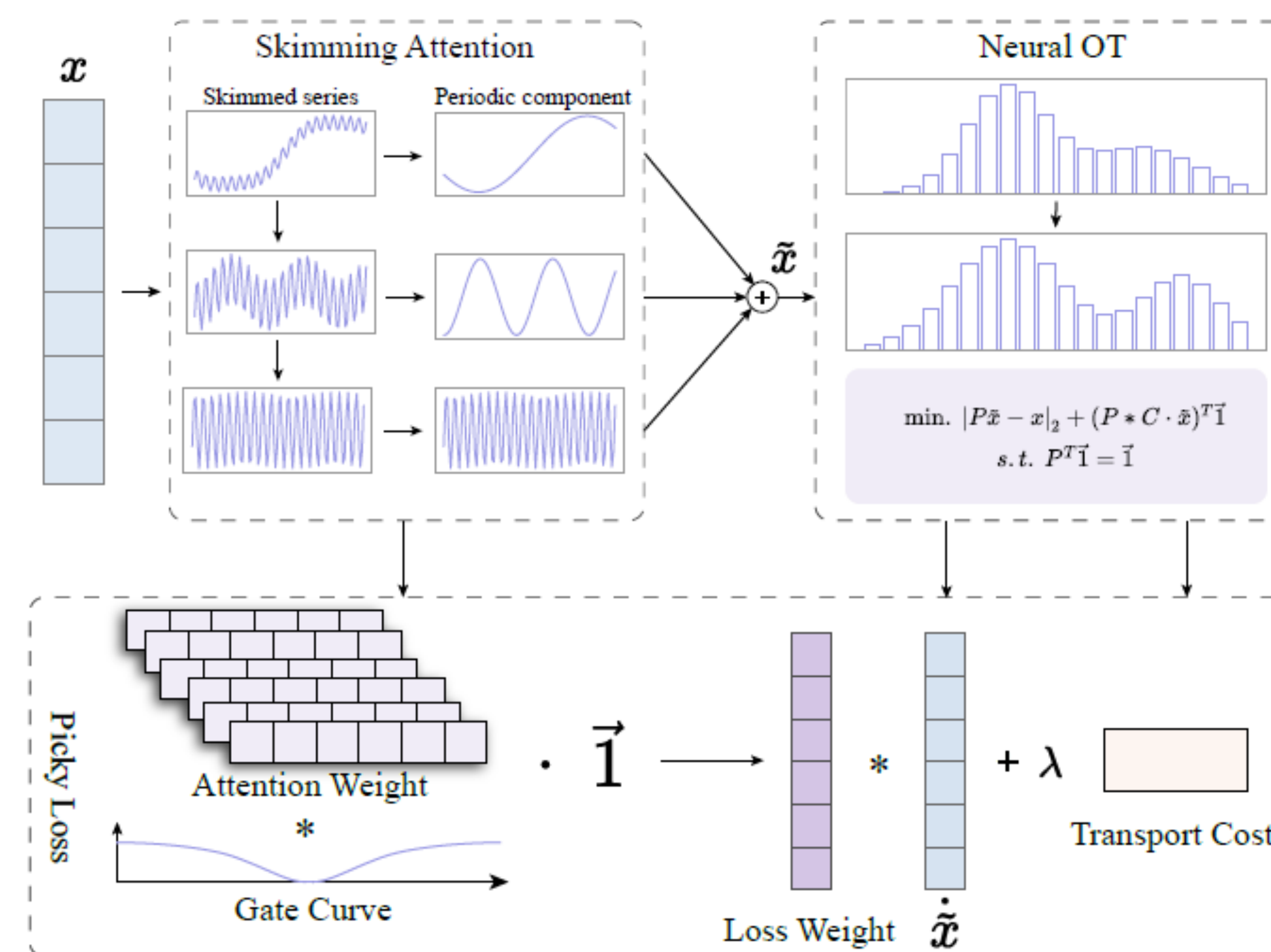
(a) The model architecture of SORN

SORN's Solution 2: Neural Optimal Transport (OT)

Goal: Filter Non-Slowing Fluctuations (Distinguishing slowdowns from speedups)

- The Solution:**

The **Neural Optimal Transport (OT)** module is used to adjust the reconstructed normal pattern, based on a custom "cost matrix" that penalizes only unwanted shifts.



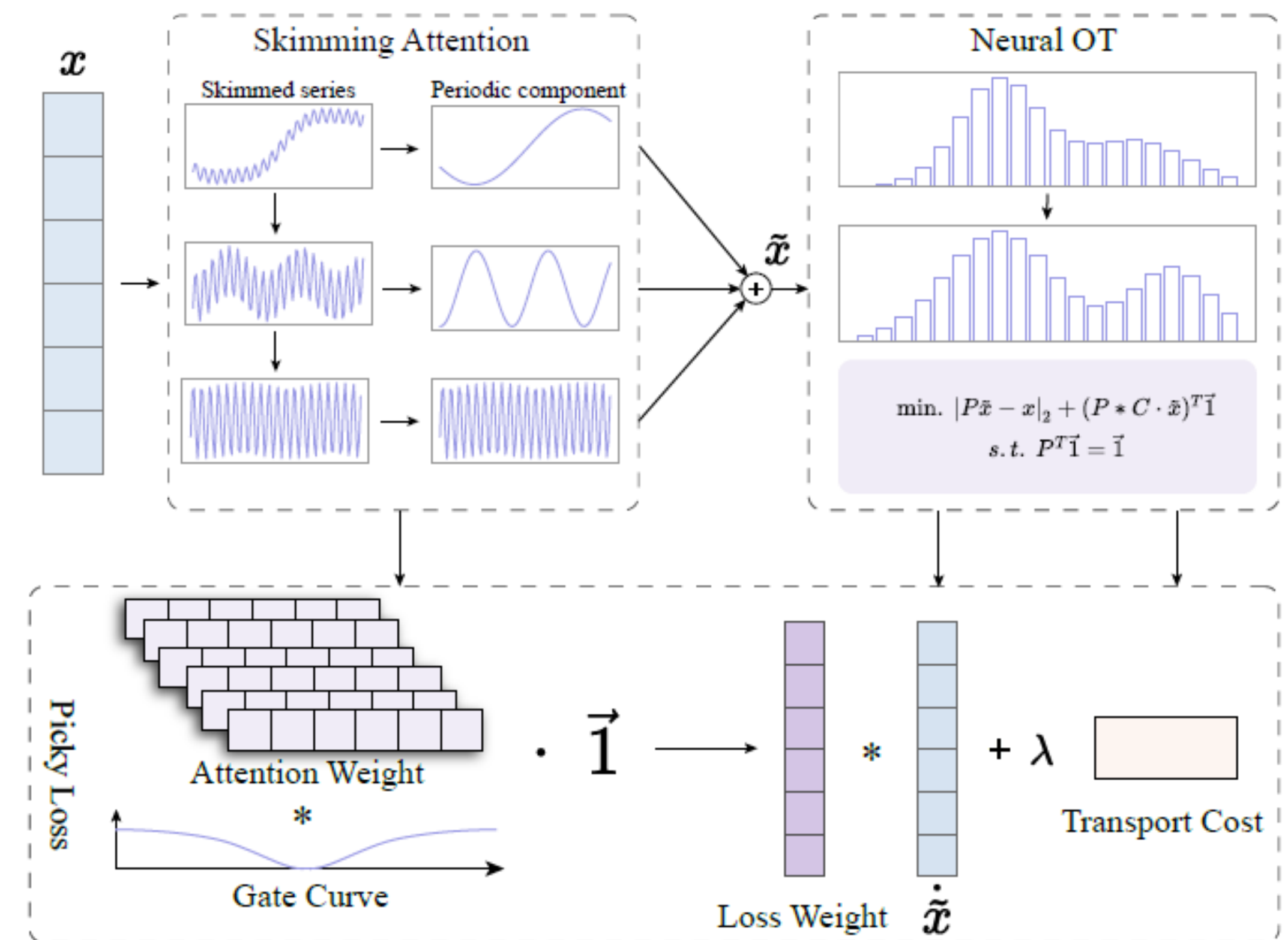
(a) The model architecture of SORN

SORN's Solution 2: Neural Optimal Transport (OT)

Goal: Filter Non-Slowing Fluctuations (Distinguishing slowdowns from speedups)

- **How it Works:**

This module is implemented as an end-to-end trainable neural network layer. The key is the custom **Transport Cost** matrix, which is engineered to assign a cost **only** if the distribution shifts towards **longer task durations (a slowdown)**.



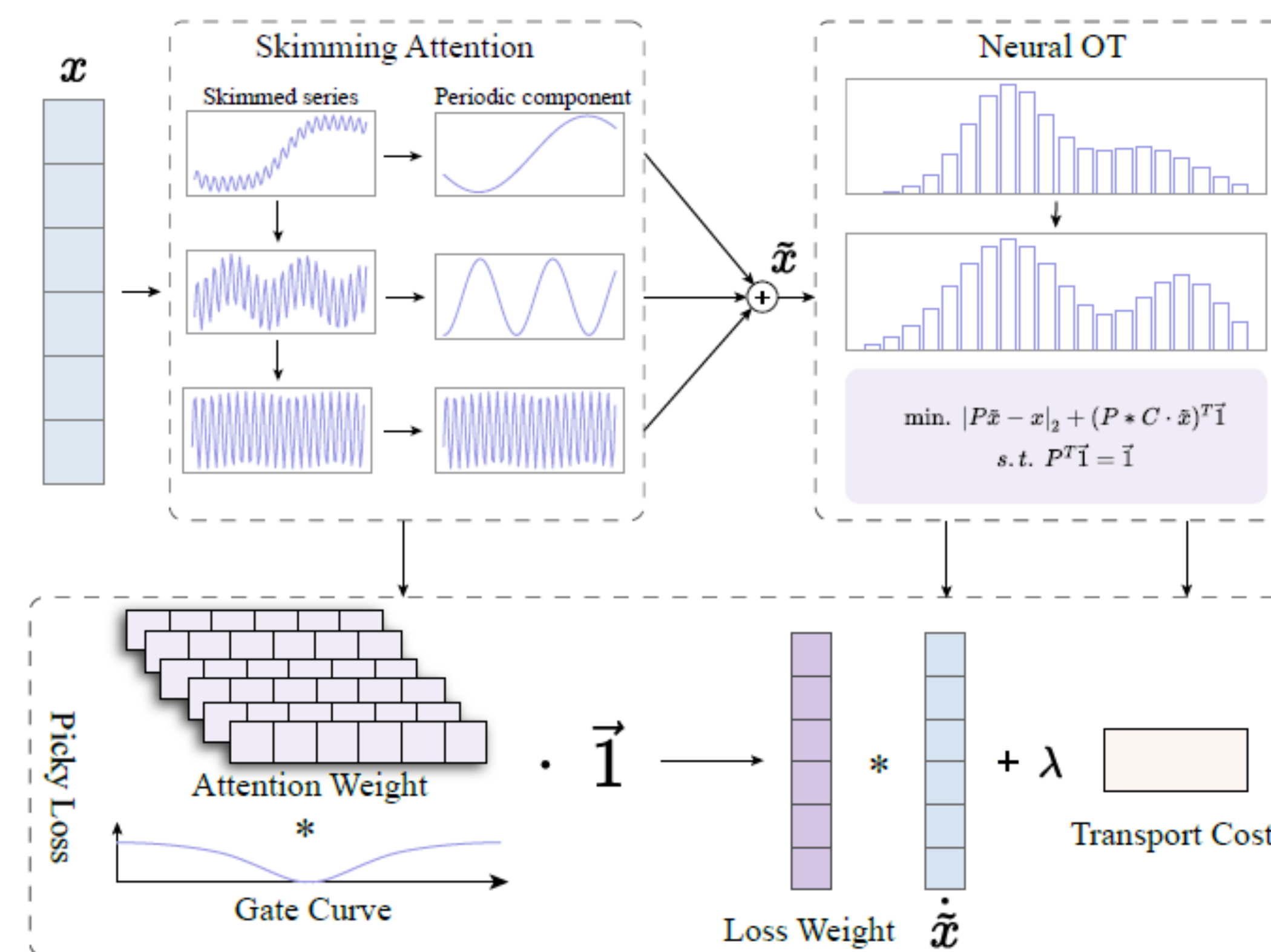
(a) The model architecture of SORN

SORN's Solution 2: Neural Optimal Transport (OT)

Goal: Filter Non-Slowing Fluctuations (Distinguishing slowdowns from speedups)

- **The Result:**

Shifts towards shorter task durations (a speedup) incur **zero cost**. This design explicitly forces the model to learn and reconstruct the speedups, ensuring that only actual cluster slowdowns trigger the anomaly alarm.



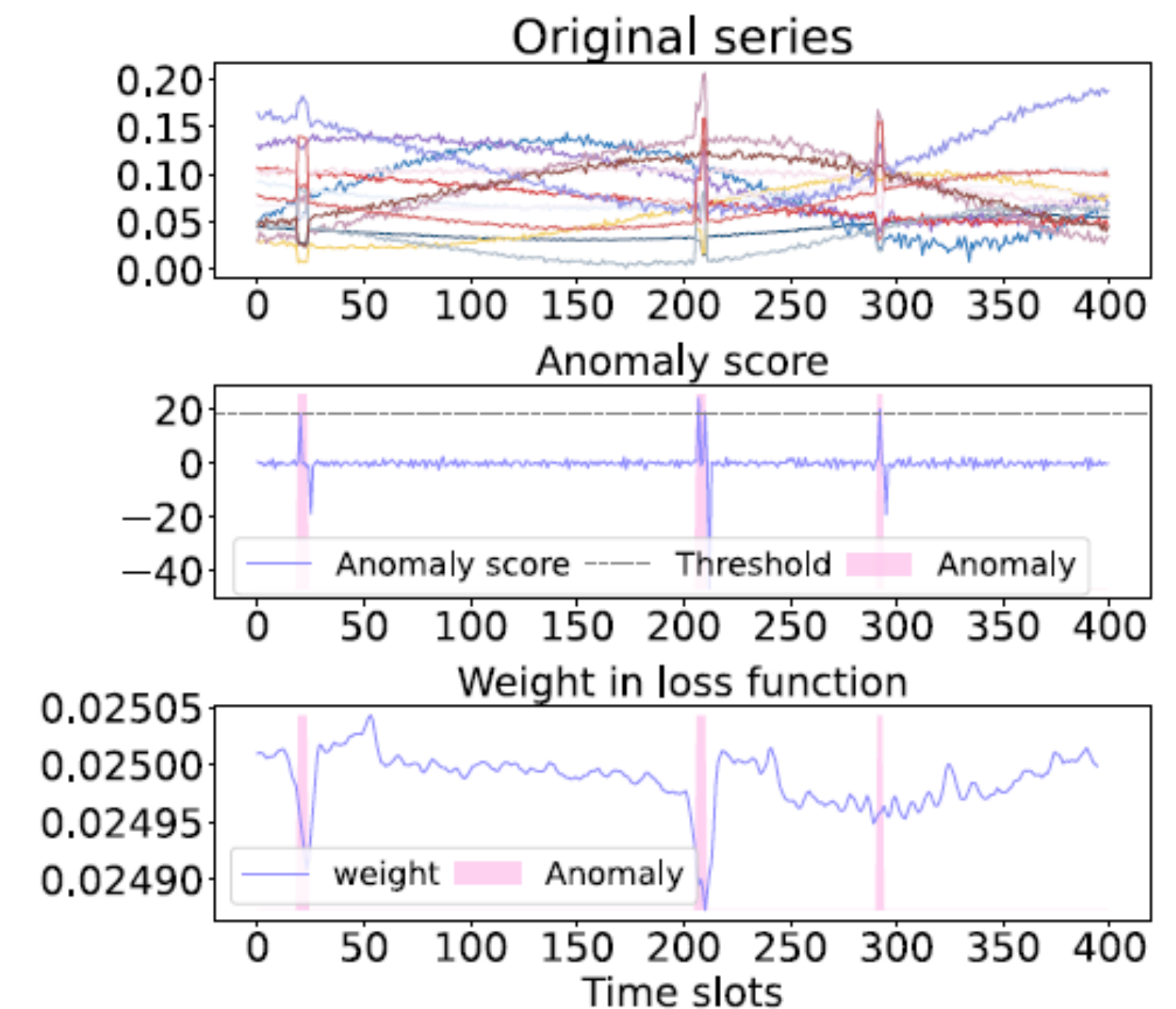
(a) The model architecture of SORN

SORN's Solution 3: Picky Loss Function

Goal: Ensure Robust Training (Mitigating training data contamination)

- **The Problem:**

Unlabeled anomalies in the training set cause the model to mistakenly learn "bad" behavior as "normal," compromising future detection accuracy.



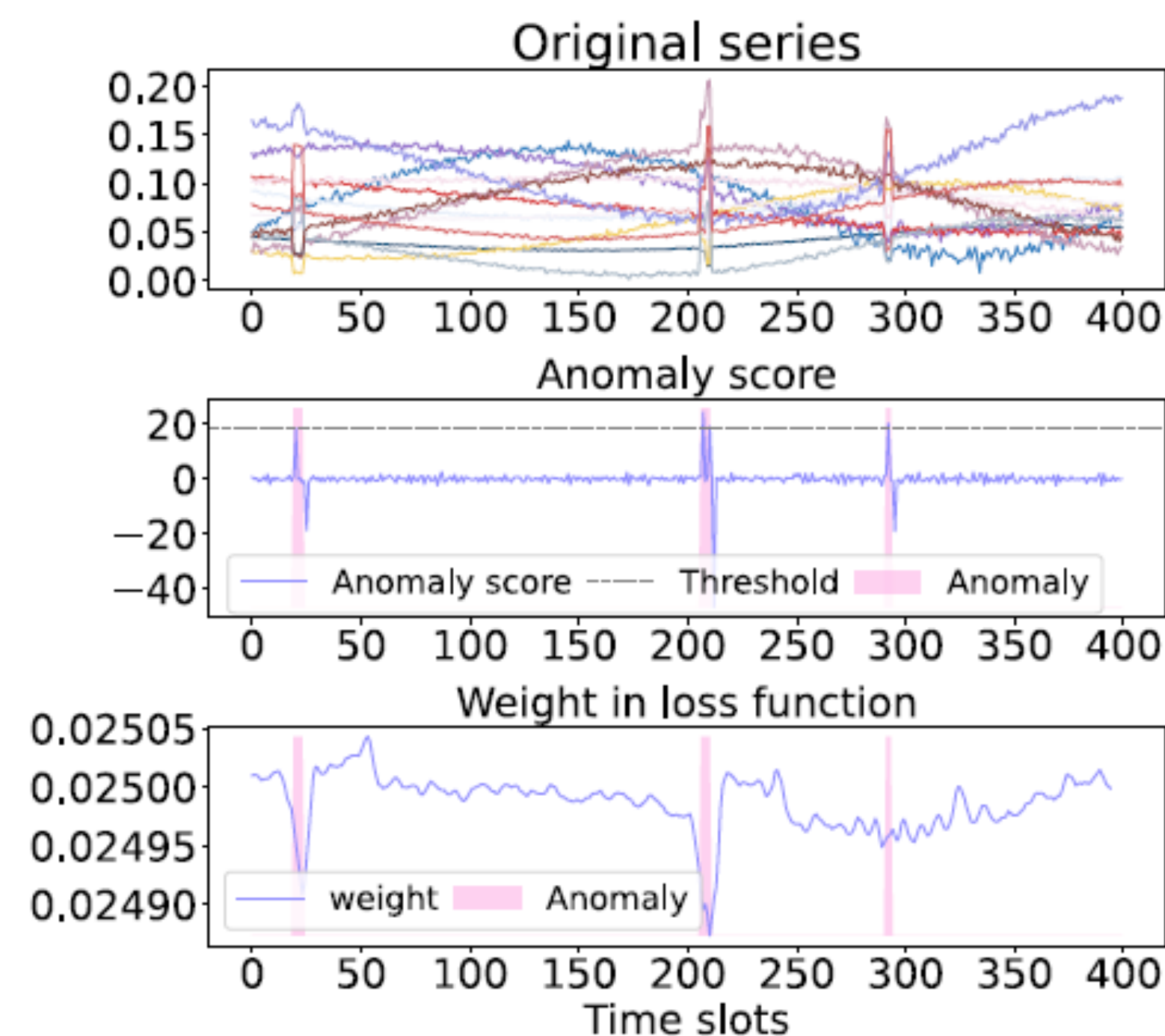
(d) The reconstruction and loss weight of SORN

SORN's Solution 3: Picky Loss Function

Goal: Ensure Robust Training (Mitigating training data contamination)

- **The Solution:**

The **Picky Loss Function** adaptively assigns a weight W to the training loss for every time slot based on its reliability.



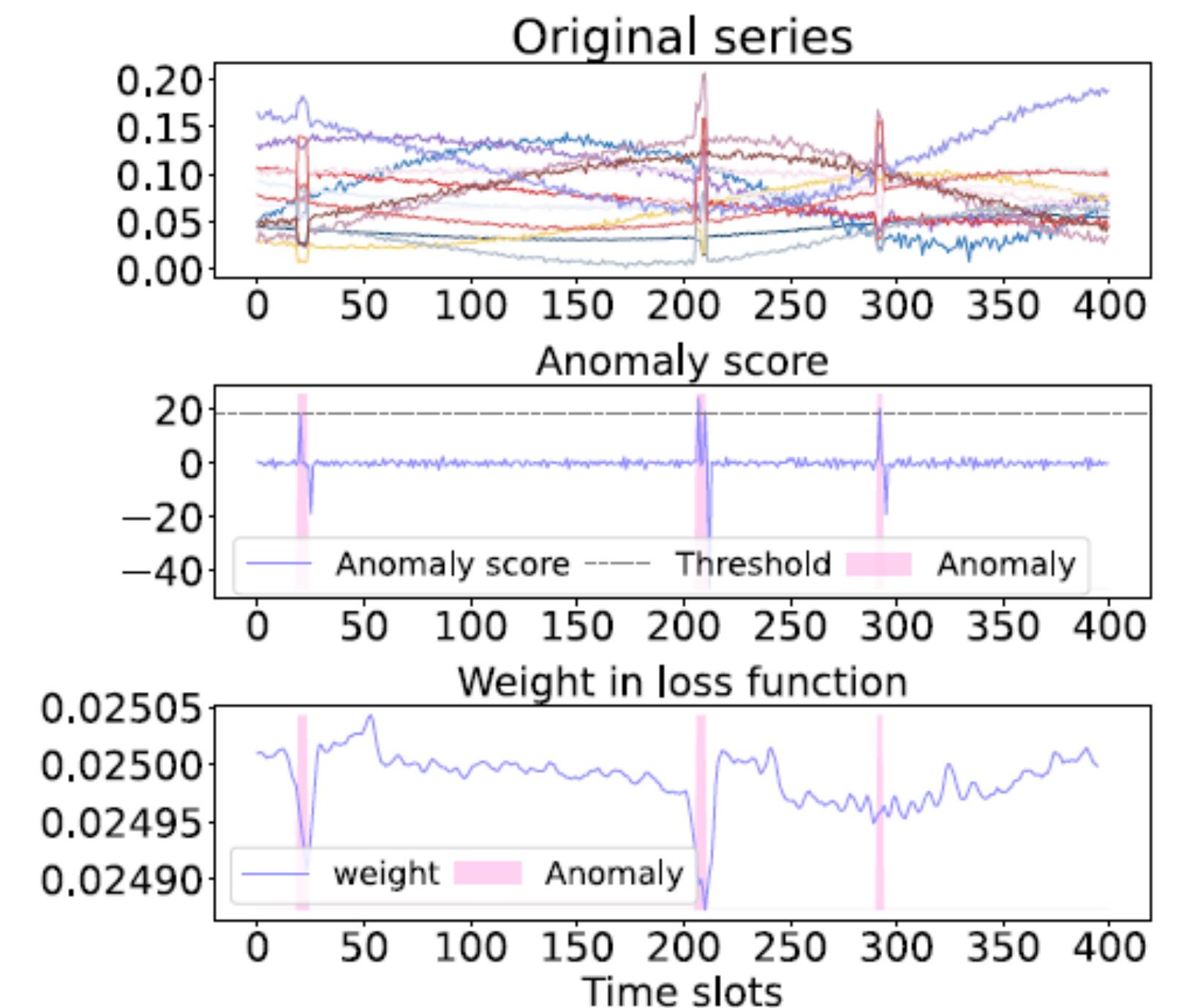
(d) The reconstruction and loss weight of SORN

SORN's Solution 3: Picky Loss Function

Goal: Ensure Robust Training (Mitigating training data contamination)

- **How it Works:**

The system uses attention patterns as a proxy for trustworthiness: **Normal points** show "wide-broad informative association" (attention spread across many periodic points), while **anomalous points** concentrate attention only locally.



(d) The reconstruction and loss weight of SORN

SORN's Solution 3: Picky Loss Function

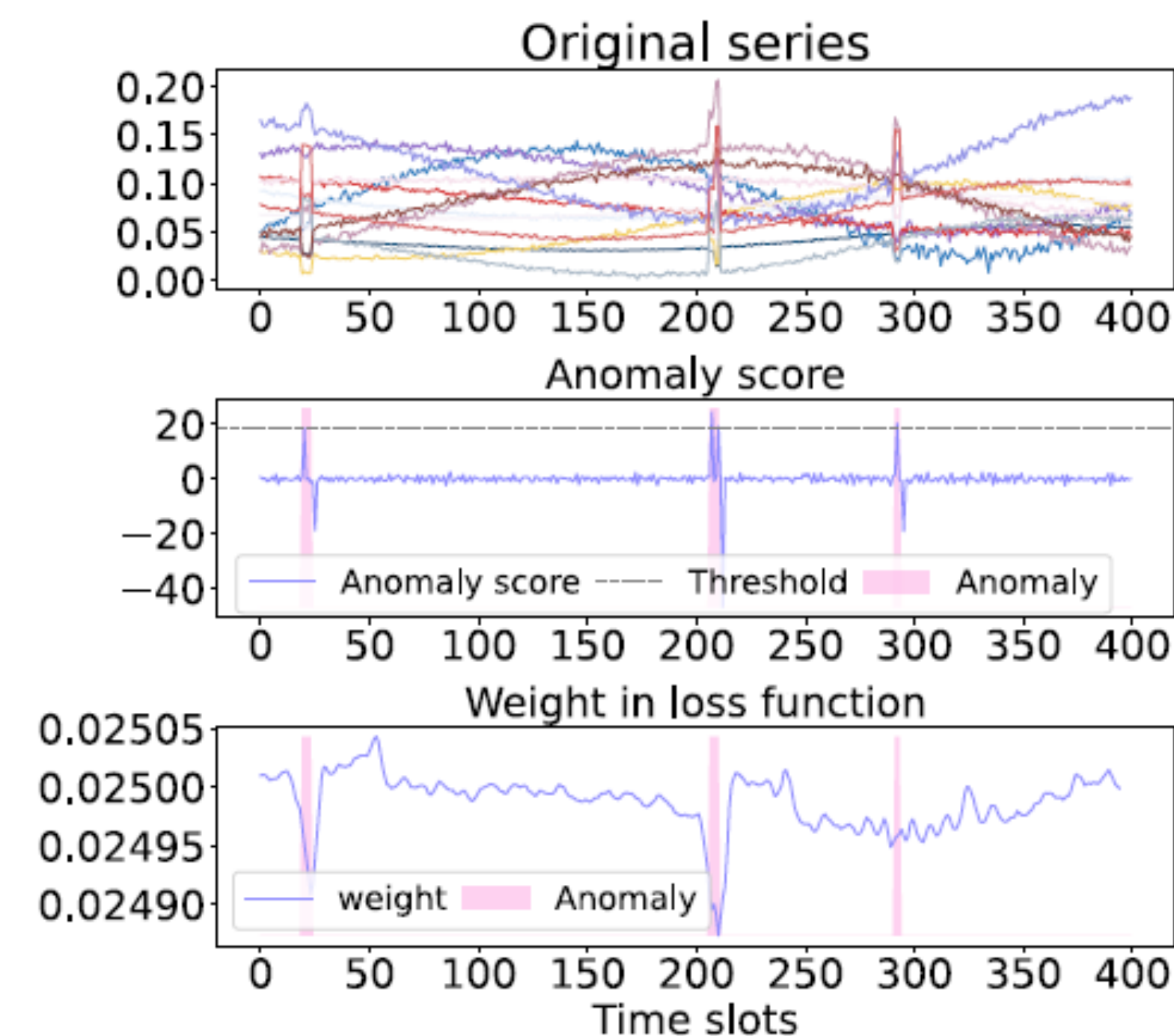
Goal: Ensure Robust Training (Mitigating training data contamination)

- **The Result:**

Time slots that are highly predictable and reliable (normal) are given a **higher weight** $W[t]$, granting them more influence during training and protecting the model from learning contaminated data.

the final loss includes the adaptive weight W :

$$\mathcal{L} = \sum_{t=0}^T W[t] (\text{Error} + \lambda \cdot \text{Cost})$$



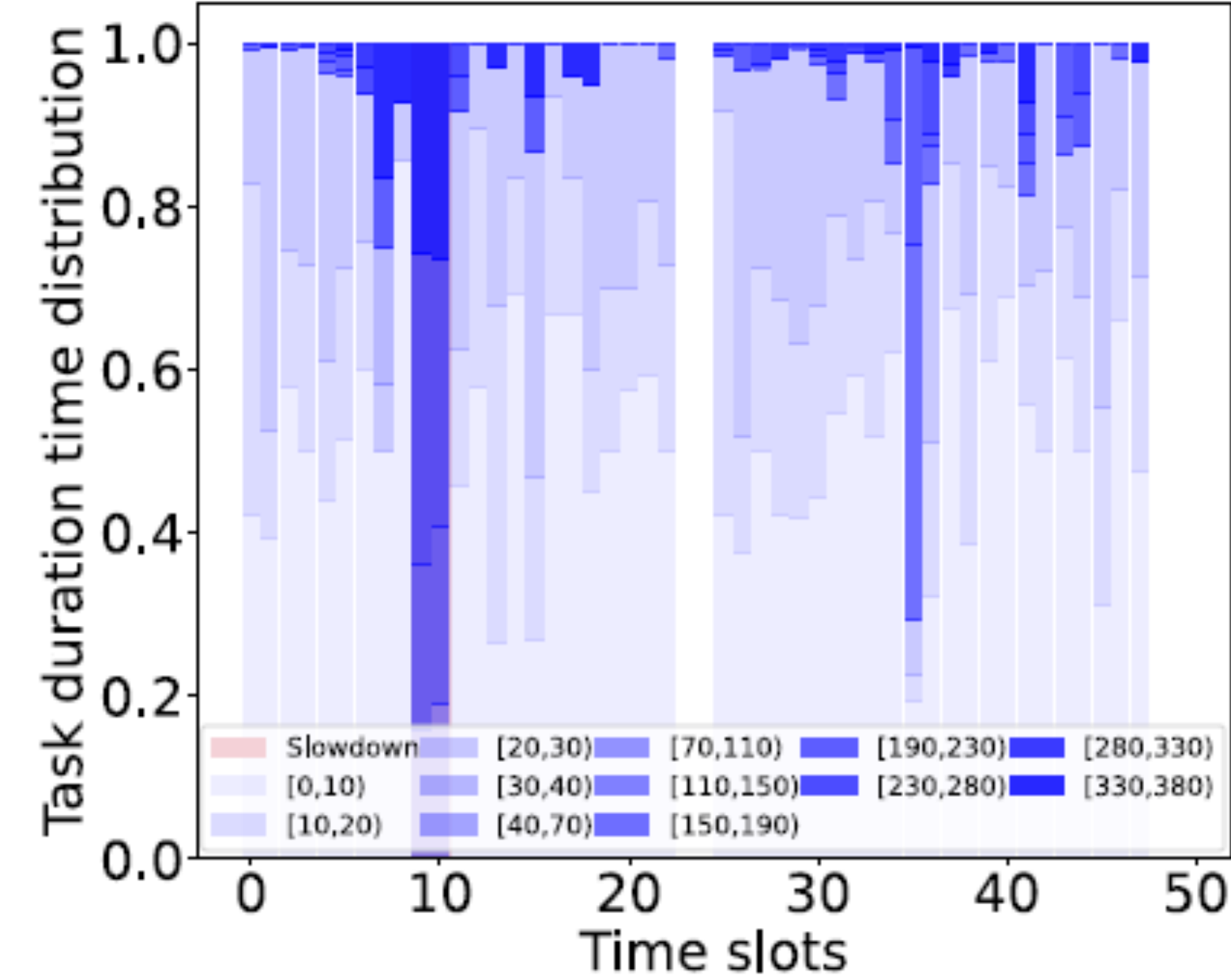
(d) The reconstruction and loss weight of SORN

The Anomaly Score

The score must quantify the operational impact:
How much slower the cluster is running.

The Metric	SORN uses an Expectation-Based Anomaly Score that measures the difference between the average expected task duration.
The Calculation	Anomaly Score = (Observed Expected Task Duration) - (Reconstructed Expected Task Duration)
The Insight	This difference is calculated using the midpoint duration $M[d]$ of each interval. A high positive score directly quantifies the magnitude of performance degradation in time , providing a critical, actionable metric for SLA management.

$$\text{AnomalyScore}[t] = \sum_{d=0}^D x[t, d] \cdot M[d] - \sum_{d=0}^D \tilde{x}[t, d] \cdot M[d]$$

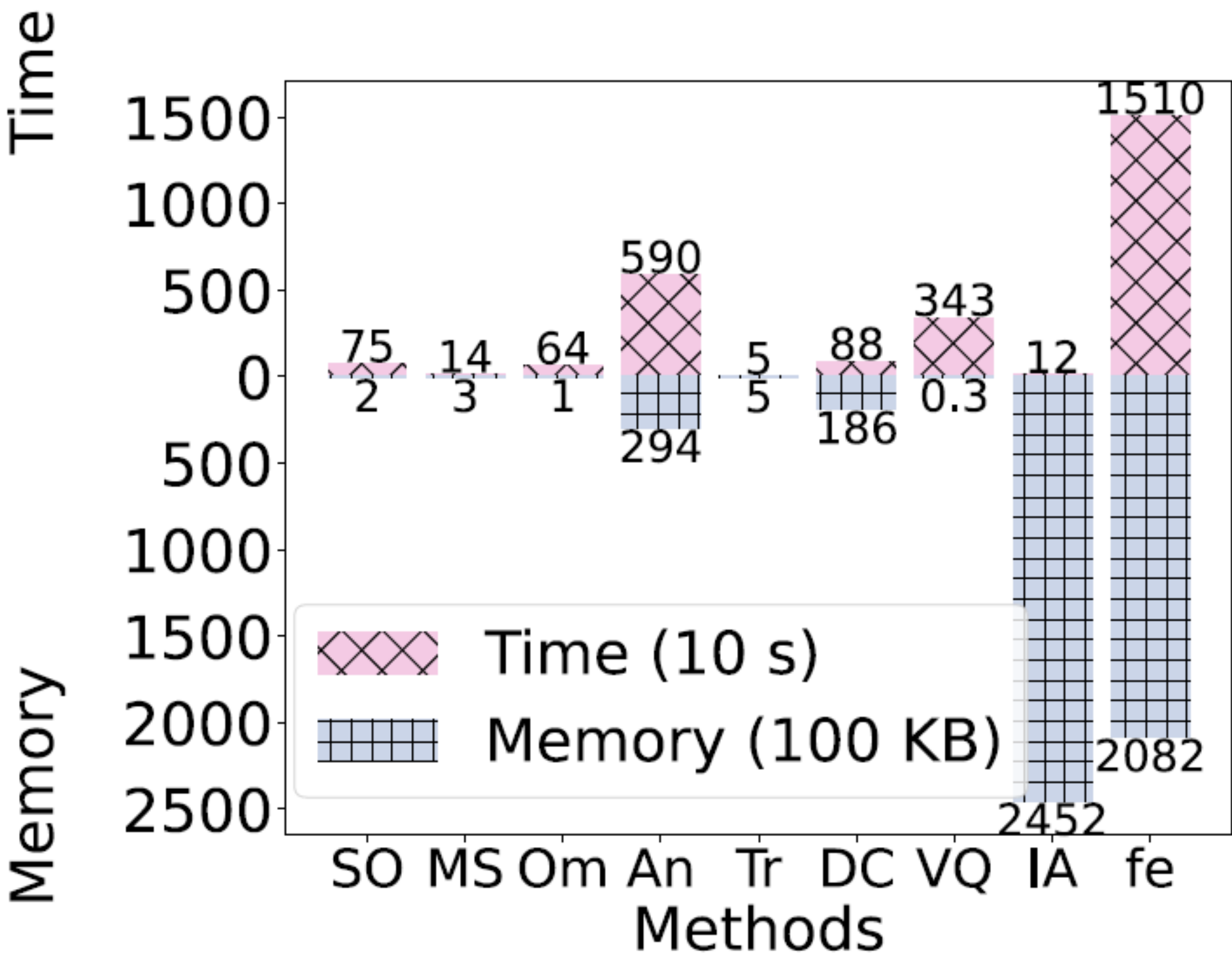


(a) The task duration time distribution

Slowdown tasks (the colored mass) shift towards the darker colors (longer duration times), increasing the **Observed Expected Duration** and resulting in a **high anomaly score**.

Results and Future Trajectories

Superior Accuracy (F1 Score)	SORN achieved the best F1 scores on all tested datasets (Ali1, Ali2, Mustang, Sync) compared to ten State-of-the-Art methods.
Operational Efficiency	SORN introduces only marginal time and memory overhead compared to light methods like <u>OmniAnomaly</u> , but achieves much better accuracy than heavier transformer-based methods.
Future Work	This robust detection system forms a foundation for advanced automation: The plan is to integrate with Large Language Models (LLMs) for deeper root cause analysis and employ Multi-Agent Systems for automatic recovery.



(b) Time and memory overhead

SORN (SO) is near the left edge (low overhead) while achieving the best overall accuracy

Future Work

SORN is highly accurate and efficient, but detection is just the first step. The future work focuses on a self-healing cloud: first, using SORN to detect the slowdown; second, using Large Language Models to analyze the root cause; and third, using Multi-Agent Systems to automatically initiate recovery.